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1. Mawaheb — DGE Talent & Career Platform

AI-powered talent development and career-enablement platform built for the UAE's Department of Government Enablement (DGE), spanning 13 use cases across candidate assessment, interviewing, coaching, and labor-market intelligence. As the sole full-stack developer, delivered the platform's core modules end-to-end: Talent Assessment with dynamic, AI-generated questionnaires (created through a live ML WebSocket "creation agent") and async media-answer evaluation; AI Interview driven by a self-hosted digital-avatar interviewer (DUIX on an Azure GPU VM, later migrated to Anam CARA-3 real-time WebRTC avatars) with Azure Speech-to-Text and a fully conversational, turn-based ML interview engine; and a Skills Observatory analytics dashboard comparing workforce demand vs. supply across emirates, industries, and roles. Architected a secure ML integration layer where the backend proxies every ML call (WebSocket + REST) so no ML credential or secret ever reaches the browser, with all configuration sourced from Azure Key Vault and short-lived OAuth2 client-credential tokens.

Tech Stack: Vue 3, TypeScript, Vite, Express 5, Knex + Objection.js (Clean Architecture), PostgreSQL on Azure, WebSocket (ws) proxies, Azure Key Vault, Azure Blob Storage, Azure Communication Services (email), Azure Speech SDK (STT), Anam CARA-3 (live avatar/WebRTC), Core42 TTS, MSAL / Azure Entra ID SSO, JWT, Hashids, Chart.js, Winston.

Impact: Unified UAE workforce development — dynamic skills assessment, avatar-led mock interviews, and labor-market analytics — into a single government platform, replacing manual assessment and interview processes with scalable, bilingual (Arabic/English) AI experiences while keeping all data and ML credentials within a UAE-hosted, Key Vault-secured backend.

2. AI Employee Agent

Multi-agent productivity platform that unifies a user's daily tools (Gmail, Outlook, Jira, GitHub, Notion, Google Drive, Google Calendar, WhatsApp) into one intelligent assistant. Built a multi-agent orchestration layer that auto-detects complex tasks and routes them to role-specific agents, plus a visual workflow editor (ReactFlow) for chaining steps. Implemented a three-tier memory architecture (Redis cache, PostgreSQL, pgvector with 3072-dim embeddings) to power RAG over user-uploaded documents and inject relevant context into agent responses. Real-time updates flow through SSE, with cron-based analyzers monitoring stale PRs, Jira blockers, calendar conflicts, and urgent emails.

Tech Stack: Next.js 16, React 19, TypeScript, NestJS 10, Python FastAPI, PostgreSQL with pgvector, Redis, Celery, OpenAI GPT-5, text-embedding-3-large (3072-dim), 32 function-calling tools, OAuth (Google, GitHub, Jira, Notion), Tailwind, shadcn/ui, ReactFlow.

Impact: Cut daily tool-switching overhead by unifying 8+ work apps into a single assistant, automating routine coordination (email triage, ticket creation, meeting prep, status alerts) and saving users multiple hours of busywork every day.

3. LMS (Learning Management System)

Full-stack, AI-powered LMS for adult professional development. Built around 27 modular NestJS services and a Vue 3 frontend, supporting students, faculty, and leadership with adaptive learning, mock interviews powered by HeyGen avatars and GPT-5, AI career counseling, automatic course generation via ML services, and skill gap analysis. Includes certificate generation with PDFKit, workforce analytics dashboards, and full multi-tenant isolation across companies, validated with 95+ unit and end-to-end tests.

Tech Stack: NestJS, Vue 3, TypeScript, PostgreSQL 16 with TypeORM, Redis, Azure Blob Storage, HeyGen API, OpenAI GPT-5, ApexCharts, Chart.js, ECharts, Tailwind, JWT auth.

Impact: Replaced traditional classroom and human-led mentoring with on-demand AI tutoring, mock interviews, and career counseling, letting organizations scale workforce upskilling to thousands of learners without growing their training team.

4. NEP (National Experts Program, UAE)

Talent assessment and career alignment platform built for the UAE's National Experts Program to evaluate candidates across AI expertise, leadership capacity, strategic thinking, growth mindset, and ethical leadership. The pipeline runs from resume upload (parsed via Affinda and a custom ML server), through a 5-question AI text interview, two phased MCQ assessments, and a final consolidated leadership score weighted across four pillars. Async background jobs handle ML server polling while Socket.io pushes real-time progress to the frontend.

Tech Stack: Express.js, TypeScript, PostgreSQL on Azure, Knex migrations, Objection.js ORM, Bull, Redis, Socket.io, OpenAI GPT-5, Affinda, custom Python ML server, Azure Blob Storage, Azure Communication Services, Puppeteer for PDF reports.

Impact: Replaced weeks of manual screening for the UAE's flagship national AI talent program with a fully automated multi-stage assessment pipeline, enabling the government to evaluate hundreds of candidates objectively and at scale.

5. Assessment Portal

Comprehensive assessment management platform for recruiting workflows that lets organizations administer and auto-evaluate English language proficiency and behavioral assessments. Candidates complete multi-section tests (MCQ, speaking, writing, reading, listening) with audio recording in the browser, while the backend pushes responses to a Python ML server that scores each section and normalizes results to a 0 to 100 scale. Recruiters get dashboards for job management, candidate tracking, and bulk assignments, with Affinda-powered resume parsing layered in for skill extraction.

Tech Stack: Vue 2, Vuetify, Vuex, Express.js, TypeScript, PostgreSQL, Objection.js, Knex, Redis, Bull, Socket.io, Azure Blob Storage, Azure Cognitive Services (Speech-to-Text), Affinda, OpenAI.

Impact: Eliminated manual grading for language and behavioral assessments, cutting candidate evaluation turnaround from several days to a few minutes and freeing recruiters to focus on shortlisting instead of scoring.

6. Recruitment Portal

Multi-tenant HR and recruitment SaaS platform that streamlines the full hiring funnel with AI-powered candidate screening, automated MCQ and coding assessments, video interview recording with transcription, and intelligent candidate ranking. GPT-5 evaluates interview responses, Affinda parses CVs into structured data, and a Python ML service handles skill extraction and job-to-candidate matching. Multi-tenancy is enforced through company-scoped middleware on every request.

Tech Stack: Express.js, TypeScript, Vue.js 2, Vuetify, PostgreSQL, Objection.js, Knex, Bull, Redis, Socket.io, OpenAI GPT-5, Affinda, Azure Blob Storage, Azure Speech Services, AWS S3, JWT with Passport (Google and LinkedIn OAuth).

Impact: Powers end-to-end hiring for 10,000+ monthly active users across multiple organizations, automating CV screening, video interview evaluation, and candidate ranking that previously consumed days of recruiter time per applicant.

7. AI Assessment Platform

An AI-powered assessment and interview platform designed to automate the hiring and evaluation process. Engineered a secure architecture where the Node.js backend acts as an intelligent proxy layer, securely authenticating and routing requests to an isolated ML server without exposing API keys to the client. Developed a real-time WebSocket proxy to power an interactive "Creation Agent" that dynamically generates customized assessments using LLMs based on natural language prompts. Implemented comprehensive result evaluation workflows, where candidate attempts are processed asynchronously to generate structured AI performance reports. Fully integrated with Azure for media storage, email communications, and speech-to-text processing.

Tech Stack: Vue 3, Vite, TypeScript, Express.js 5, Knex, Objection.js, PostgreSQL, WebSockets (ws), Bull, Redis, Azure (Blob Storage, Communication Services, Speech SDK, Key Vault), MSAL (Entra ID SSO).

Impact: Automated the end-to-end assessment pipeline—from dynamic test generation to AI-evaluated results—drastically reducing the time recruiters spend creating questionnaires and manually grading candidate attempts.

8. Transportation Platform (Multi-Tenant SaaS for UK Transport Operators)

Architected a Turborepo monorepo housing 5 applications: a Transport Management System, a Warehouse Management System, and dedicated mobile apps for drivers, technicians, and customers. Built a comprehensive job management flow, automated Stripe invoicing, a finance system with PDF generation, and a Workshop Management System with offline-first mobile inspections. Fleet management includes real-time tracking, with 10+ third-party integrations layered in.

Tech Stack: Next.js 16, React 19, TypeScript, Convex BaaS, Drizzle ORM, PostgreSQL, React Native (Expo 54), Turborepo, TanStack Query, Stripe, Mapbox, Playwright.

Impact: Achieved 99% uptime and 35% faster API response times.

9. UK Taxi Management System

End-to-end taxi management platform featuring an intelligent dynamic pricing algorithm, an AI-powered dispatch system that optimizes driver assignment, and a secure payment layer integrating Stripe and Wonderful Payments with PCI-DSS compliance. Added real-time driver tracking, automated email notifications, and a multi-app frontend covering operators, drivers, and customers.

Tech Stack: PHP, Laravel, React.js, Vue.js, Next.js, Nuxt.js, TypeScript, MySQL, Redis, Stripe API.

Impact: 20% revenue increase from intelligent pricing and 30% faster wait times via AI dispatch.

10. Enterprise ERP System

Multi-tenant ERP serving 5 housing societies with 3,000+ residents and 1,000+ employees. Built 12 integrated modules including HRMS, a 5-level Chart of Accounts with double-entry ledger, a full procurement cycle (demand to quotation to PO to GRN to invoice to payment), inventory management with FIFO valuation and barcode support, project management, and real estate units management. Real-time updates flow through Socket.io and Pusher.

Tech Stack: Laravel, Vue.js, SQL Server (Multi-DB), OAuth2 with Passport, Spatie Permission, Socket.io, Laravel Echo, Pusher, DomPDF, Maatwebsite Excel.

Impact: Reduced manual work by 70% across daily operations.

11. POS (Triangle POS for Automotive Shops)

Modular point-of-sale system tailored for automotive service shops and retail businesses. Built a real-time Livewire-driven checkout with cart management, tax and discount handling, and partial payments, alongside inventory tracking with barcode support, stock alerts, and multi-unit pricing. Includes 6 detailed reports (Profit/Loss, Sales, Purchases, Returns, Payments) and Twilio-based WhatsApp and SMS notifications for service reminders.

Tech Stack: PHP, Laravel, Livewire, MySQL, Bootstrap with CoreUI, Spatie Permission and MediaLibrary, DomPDF, Maatwebsite Excel, Twilio SDK, Milon Barcode.

Impact: Replaced paper-based billing and manual stock tracking for automotive shops, giving owners real-time visibility into sales, inventory, and customer service reminders from a single dashboard and significantly reducing daily bookkeeping effort.

12. HR & Payroll Management System

End-to-end HRMS designed for organizations with 1,000+ employees. Features automated attendance tracking via biometric integration, leave management with accrual rules, full payroll processing covering overtime, deductions, and allowances, a built-in recruitment portal with interview scheduling, loans and advances, final settlement calculations, and 12+ analytics dashboards.

Tech Stack: Laravel, Vue.js, SQL Server, T-SQL stored procedures, ZKTeco SDK for biometrics, ApexCharts, Chart.js.

Impact: Reduced payroll processing time by 60% and eliminated end-of-month manual reconciliation across HR and finance teams.

13. Airbooking System

Online travel booking platform for flights and hotels, built end-to-end for an SA Systems client. Sole developer on the project — built the complete Nuxt.js frontend and Laravel backend, integrated a client-provided third-party travel API for live flight and hotel inventory and search, and implemented the full booking flow with secure card payments via Stripe.

Tech Stack: Nuxt.js, Vue.js, Laravel, PHP, MySQL, Stripe API, REST APIs, third-party flights/hotels API.

Impact: Enabled the client to offer self-serve online flight and hotel booking with real-time availability and integrated card payments, handling the full journey from search to paid booking without manual intervention.

14. Exclusive Customs — E-commerce Platform

E-commerce platform for a UK car-modification retailer (exclusivecustoms.co.uk), built as a ground-up replacement for the client's unstable WordPress store, which had a dated UI and checkout issues that were costing them orders. Sole full-stack developer — built the storefront and backend entirely in Next.js (Node.js API routes) with MySQL, implemented PayPal checkout alongside a bank-transfer payment option, and delivered a dedicated dealer portal with trade pricing, bulk ordering, and order tracking.

Tech Stack: Next.js, React, Node.js, MySQL, PayPal API, REST APIs.

Impact: Replaced a failing WordPress site with a stable custom storefront, resolving the order-placement issues customers were hitting, and opened a trade sales channel through the dealer portal — retail and dealer orders now run through one platform.

15. BreeO — Brand & E-commerce Platform

Brand and e-commerce platform for BreeO, a multi-enzyme detergent brand by Olympia Chemicals, Lahore. Sole full-stack developer — built the React frontend and Laravel backend with MySQL, covering the product catalog (fabric-care and home-care lines), bundle deals, customer registration and accounts, ordering with both online payment and cash-on-delivery, newsletter signups, and a content hub spanning laundry-tip guides, blogs, FAQs, press releases, and careers. Also built the complete admin panel for managing products, orders, blogs, and site content.

Tech Stack: React, Laravel, PHP, MySQL, REST APIs.

Impact: Gave the brand a full direct-to-consumer channel — online ordering with card and COD payments plus content marketing — alongside its traditional retail distribution, all manageable by the client from a single admin panel.

16. SA Systems — Official Company Website

Complete revamp of SA Systems' official corporate website, replacing the company's legacy WordPress site with a custom-built platform. Sole full-stack developer — built the Next.js frontend and Laravel backend with MySQL, covering service and product pages, portfolio, an Insights/blog section, careers, newsletter, and multi-office contact. Implemented SEO optimization across the site — server-rendered pages, meta management, Google Tag Manager and Search Console integration — and built an admin panel from which all frontend content (services, products, insights, and site copy) is managed without developer involvement.

Tech Stack: React, Laravel, PHP, MySQL, REST APIs, Google Tag Manager.

Impact: Moved the company's primary marketing and lead-generation channel off WordPress onto a fully owned, SEO-optimized stack — the site still runs on this build today, with all content managed in-house through the custom admin.

17. SA Gardens — Housing Society Website with 3D Virtual Tour

Public showcase website for SA Gardens, a Lahore housing society (parent company of SA Systems), presenting the society's projects, payment plans, and its centerpiece feature: an interactive 3D virtual tour of the entire society, rendered in-browser. Sole frontend developer — built the complete React application and the WebGL tour using react-three-fiber and drei, assembling client-provided 3D assets into a navigable virtual experience that lets prospective buyers explore the society's layout remotely instead of relying on brochures or physical site visits.

Tech Stack: React, Three.js, react-three-fiber, drei, JavaScript, WebGL, HTML/CSS.

Impact: Gave the society a differentiated digital sales tool — buyers could virtually walk the society before visiting, moving property marketing beyond static maps and printed payment plans.

18. Stond — Tiles & Flooring Company Website

Static showcase website for Stond, a home tiles and flooring company (a sister venture of SA Gardens), built during the SA Systems engagement. Sole frontend developer — built the complete React site with Bootstrap, presenting the company's product range and business information. The site has since been retired.

Tech Stack: React, Bootstrap, JavaScript, HTML/CSS.

Impact: Gave the company its first dedicated digital presence for showcasing its tile and flooring range to prospective buyers.